Unilateral z-transform and mixed practice

Practice questions · Day 16 · Sun 8 Nov · pack D9

5. Practice

From your course files

T3 Q3 (tutorial; do (b), redone with $Y = HX$). A LSI system is described by the LCCDE

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n] - x[n-1].$$

The system is initially at rest, i.e. $y[n] = 0$ for $n < 0$.

(a) Write down the characteristic modes of this system.
(b) Find the output of the system if the input signal is $x[n] = 3^{-n}u[n]$.

T3 Q5 (tutorial; do (b) and (c) via $H(z)$). Determine the impulse response $h[n]$ of the following DT LSI systems:

(b) $y[n] + 0.2y[n-1] - 0.08y[n-2] = x[n]$
(c) $y[n] - 2y[n-1] + y[n-2] = x[n]$

T7 Q1 (tutorial). Determine all possible signals $x(n)$ with the following z-transform:

$$X(z) = \frac{5z^{-1}}{(1-2z^{-1})(3-z^{-1})}$$

T7 Q2 (tutorial). Determine the signal $x(n)$ with the z-transform:

$$X(z) = \frac{3}{1 - \frac{10}{3}z^{-1} + z^{-2}}$$

if $x(n)$ is absolutely summable.

T7 Q3 (tutorial). A *causal* discrete-time LTI system produces the output

$$y(n) = \left(\frac{1}{3}\right)^n u(n)$$

in response to the input signal

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4}\left(\frac{1}{2}\right)^{n-1} u(n-1).$$

(a) Determine the impulse response $h(n)$ and transfer function $H(z)$ of this system.
(b) Find the difference equation that characterises the system.
(c) Is this system stable?

New questions

Part A: unilateral z-transform (graded easy to hard)

Q1 [4 marks]. $y(n) - 0.5y(n-1) = x(n)$ with $x(n) = u(n)$ and $y(-1) = 4$.

a. Find $Y(z)$. [2] (b) Find $y(n)$. [1] (c) Give $y_{zi}(n)$ and $y_{zs}(n)$. [1]

Q2 [5 marks]. $y(n) - 0.7y(n-1) + 0.1y(n-2) = x(n)$ with $y(-1) = 1$, $y(-2) = 5$ and $x(n) = \delta(n)$.

a. Find the IC polynomial $I(z)$. [1] (b) Find $y_{zi}(n)$. [1] (c) Find $y_{zs}(n)$ and say why it equals $h(n)$. [2] (d) Find $y(n)$ and check $y(0)$ and $y(1)$. [1]

Q3 [5 marks]. $y(n) - 0.25y(n-2) = x(n) + x(n-1)$ with $x(n) = (\frac{1}{3})^n u(n)$, $y(-1) = 2$ and $y(-2) = -4$. Find $y_{zi}(n)$, $y_{zs}(n)$ and $y(n)$.

Q4 [6 marks]. $y(n) - y(n-1) + 0.25y(n-2) = x(n)$ with $x(n) = u(n)$, $y(-1) = 2$ and $y(-2) = 0$.

a. Find $y_{zi}(n)$. [2] (b) Find $y_{zs}(n)$. [2] (c) Find $y(n)$ and explain what happened to the $n(0.5)^n$ terms. [1] (d) Check $y(0)$ and $y(1)$ by recursion. [1]

Q5 [6 marks]. $y(n) - 0.9y(n-1) + 0.2y(n-2) = x(n)$ with $x(n) = u(n)$, $y(-2) = 0$ and $y(-1) = c$.

a. Find $H(z)$, its poles and the steady-state output. [2] (b) Find $Y(z)$ in terms of c. [2] (c) Find c so that the $(0.5)^n$ mode does not appear in $y(n)$. [1] (d) Find $y(n)$ for that c. [1]

Q6 [8 marks] (chained). A causal system is described by $y(n) - 0.25y(n-1) - 0.125y(n-2) = x(n) - x(n-1)$. The input is $x(n) = (0.5)^n u(n)$ and the ICs are $y(-1) = 4$, $y(-2) = -8$.

a. Find $H(z)$, its poles and zeros, and state whether the system is stable. [2]
b. Find $y_{zi}(n)$. [2]
c. Find $y_{zs}(n)$. Explain why a term in $n(0.5)^n$ appears. [3]
d. Give $y(n)$ and check $y(0)$ and $y(1)$ by recursion. [1]

Part B: mixed z-domain set (exam-style, about 12 min each)

Q7 [8 marks] ROC and inverse. Let $X(z) = \dfrac{2 - 2.5z^{-1}}{(1 - 0.5z^{-1})(1 + 2z^{-1})}$.

a. Find the poles and zeros, including any at $z = 0$. [1]
b. Expand $X(z)$ in partial fractions. [2]
c. List every possible ROC and the matching $x(n)$, and say whether each is causal, anti-causal or two-sided. [3]
d. Which $x(n)$ has a DTFT? Evaluate that DTFT at $\omega = 0$. [1]
e. Find the unilateral transform $X^+(z)$ of the signal in (d). Why is it not equal to $X(z)$? [1]

Q8 [8 marks] Transfer function, stability and the response with ICs. A causal system is described by $y(n) + 0.1y(n-1) - 0.3y(n-2) = x(n)$.

a. Find $H(z)$, its poles and $h(n)$. [3]
b. Is the system stable? Give a reason. [1]
c. With $x(n) = u(n)$, $y(-1) = 1$ and $y(-2) = 0$, find $y_{zi}(n)$, $y_{zs}(n)$ and $y(n)$. [3]

d. What value does $y(n)$ settle to, and why do the ICs not affect it? [1]

Q9 [8 marks] Identification, then ICs. A causal LTI system, initially at rest, gives the output $y(n) = \left[4 - 4(0.5)^n + (0.25)^n\right]u(n)$ for the input $x(n) = u(n)$.

a. Find $H(z)$. [2] (b) Find the LCCDE. [1] (c) Find $h(n)$ and state whether the system is stable. [2]
b. The same system now has $y(-1) = y(-2) = 4$, as if its output had already settled at 4. It gets the same step $u(n)$ from $n = 0$. Find $y(n)$ and explain why it is not constant. [3]

Q10 [9 marks] Unstable poles, ROC and why ICs cannot fix them. Let

$$H(z) = \frac{1.5z^{-1}}{(1 - 0.5z^{-1})(1 - 2z^{-1})}.$$

a. Find the poles and zeros, including those at 0 and ∞. [1]
b. For each possible ROC, give $h(n)$ and state whether the system is causal and whether it is stable. [3]
c. Write the LCCDE of the causal system. [1]
d. Find the step response of the causal system at rest. [1]
e. Take $y(-2) = 0$. Find the $y(-1)$ that makes the step response bounded, and give $y(n)$. [2]
f. Explain in one or two sentences why (e) is not a way to build a stable system. [1]

6. Quick recall check

1. Write $\mathcal{Z}^+\{y(n-2)\}$ in terms of $Y(z)$ and the ICs.
2. The input is $x(n) = u(n)$. What is $\mathcal{Z}^+\{x(n-1)\}$?
3. Which part of the total response equals $H(z)X(z)$, and which poles can appear in the other part?
4. Which time sequence has the transform $\frac{1}{(1-pz^{-1})^2}$?
5. $Y(z) = \frac{3 - z^{-1} + 6z^{-2}}{(1-z^{-1})(1-2z^{-1})(1+3z^{-1})}$. What is $y(0)$, with no partial fractions?